

References

- [1] Helena Molina Abril. Homological spanning forests for discrete objects. *Universidad de Sevilla, PhD thesis*, 2012.
- [2] S. Aofi Al-Akbi and S. Hadi Jafari. Finite split metacyclic groups and their 2-nilpotent multipliers. 2023.
- [3] Nisreen Alokbi and Ali Sapeeh Mayah, Faik and Rashid. New invariant quantity to measure the entanglement in the braids. *J. Nig. Soc. Phys. Sci.*, 4, 1051, 2022.
- [4] V. Alvarez, J.A. Armario, M.D. Frau, and P. Real. Calculating cocyclic hadamard matrices in mathematica: Exhaustive and heuristic searches. In Andres Iglesias and Nobuki Takayama, editors, *Mathematical Software - ICMS 2006*, volume 4151 of *Lecture Notes in Computer Science*, pages 419–422. Springer Berlin Heidelberg, 2006.
- [5] Khaled Alzobydi and Graham Ellis. Cellular approximations to the diagonal map. *Mathematics of Computation*, 94(352):953 – 1002, 2025. Cited by: 0.
- [6] N. Andruskiewitsch, C. Galindo, and M. Müller. Examples of finite-dimensional Hopf algebras with the dual Chevalley property. *ArXiv e-prints*, September 2015.
- [7] Andres Angel, Eric Samperton, Carlos Segovia, and Bernardo Uribe. Oriented and unitary equivariant bordism of surfaces. *Algebraic and Geometric Topology*, 24:1623–1654, 2024.
- [8] Z. Arvasi and A. Odabas. Computing exterior isoclinism of crossed modules. *Turkish Journal of Mathematics*, 48; No. 6, Article 3, 2024.
- [9] Zekeriya Arvasi, Alper Odabaş, and Christopher D. Wensley. Computing 3-Dimensional Groups : Crossed Squares and Cat^2 -Groups. *arXiv e-prints*, page arXiv:1911.12799, Nov 2019.
- [10] Anh Tuan B and Anh Thi N. A brief introduction to the Quillen conjecture. *Sci. Tech. Dev. J.*, 22(2):235–238, 2019.
- [11] J.S. Bailes. *Cohen-Lenstra Heuristics and the Cohomology of the Braid Group*. PhD thesis, University of Melbourne. University of Melbourne, 2011.
- [12] Mohamed Barakat. The homomorphism theorem and effective computations. *Habilitationsschrift, Aachen* <http://www.mathb.rwth-aachen.de/~barakat/habil/habil.pdf>, 2018.
- [13] Ethan Berkove and Alexander D. Rahm. The mod 2 cohomology rings of SL_2 of the imaginary quadratic integers. *J. Pure Appl. Algebra*, 220(3):944–975, 2016. With an appendix by Aurel Page.

- [14] Anna M. Bigatti and E. Sáenz-de Cabezón. Computation of the $(n - 1)$ -st Koszul homology of monomial ideals and related algorithms. In *ISSAC 2009—Proceedings of the 2009 International Symposium on Symbolic and Algebraic Computation*, pages 31–37. ACM, New York, 2009.
- [15] Laura Bonn and Sebastian Giersbach. Simple totally disconnected locally compact groups separated by finiteness properties, 2026.
- [16] L.R. Bosko. *Schur Multipliers of Nilpotent Lie Algebras*. BiblioBazaar, 2012.
- [17] Rachel Boyd. Erratum to: Low-dimensional homology of finite-rank coxeter groups. <https://guests.mpim-bonn.mpg.de/rachaelboyd/Erratum.pdf>, 2021.
- [18] Piotr Brendel, Pawel Dlotko, Graham Ellis, Mateusz Juda, and Marian Mrozek. Computing fundamental groups from point clouds. *Applicable Algebra in Engineering, Communication and Computing*, 26(1-2):27–48, 2015.
- [19] A. T. Bui and Graham Ellis. Computing bredon homology of groups. *Journal of Homotopy and Related Structures*, 11(4):715–734, 2016.
- [20] Anh Tuan Bui, Alexander D. Rahm, and Matthias Wendt. On Farrell-Tate cohomology of GL_3 over rings of quadratic integers. *J. Algebra*, 615:328–357, 2023.
- [21] A.T. Bui, A. Rahm, and M. Wendt. The Farrell-Tate and Bredon homology for $PSL_4(\mathbb{Z})$ and other arithmetic groups. *ArXiv e-prints*, November 2016.
- [22] M. Bulcan, L. Li, and R. Radhakrishnan. $a \times b = c$ in $2 + 1d$ tqft. *arXiv e-prints*, page arXiv:2012.14689, Dec 2020.
- [23] Martin Čadek, Marek Krčál, Jiří Matoušek, Lukáš Vokřínek, and Uli Wagner. Polynomial-time computation of homotopy groups and Postnikov systems in fixed dimension. *SIAM J. Comput.*, 43(5):1728–1780, 2014.
- [24] Filippo Callegaro and Ivan Marin. Homology computations for complex braid groups. *arXiv*, 1011.4375, 2010.
- [25] Peter J. Cameron. Graphs defined on groups. *arXiv*, 2102.11177, 2021.
- [26] Alvaro Torass Casas. Persistence spectral sequences. *PhD thesis, Cardiff University*, 2022.
- [27] Manuela Cerdeiro. Un nuevo enfoque sobre la conjetura de whitehead y la asfericidad delos complejos lot. *PhD thesis, Universidad de Buenos Aires en el area Ciencias Matematicas*, 2015.

- [28] Erin Wolf Chambers and Mikael Vejdemo-Johansson. Computing minimum area homologies. *Computer Graphics Forum*, 34(6):13–21, 2015.
- [29] Jacopo G. Chen. Computing the twisted l^2 -euler characteristic. 2024.
- [30] Jacopo G. Chen. Some closed hyperbolic 5-manifolds, 2025.
- [31] Miranda C. N. Cheng, Paul de Lange, and Daniel P. Z. Whalen. Generalised umbral moonshine. *SIGMA Symmetry Integrability Geom. Methods Appl.*, 15:Paper No. 014, 27, 2019.
- [32] Rocco Chirivì and Mauro Spreafico. Space forms and group resolutions: the tetrahedral family. *J. Algebra*, 510:52–97, 2018.
- [33] Maura Clancy and Graham Ellis. Homology of some Artin and twisted Artin groups. *J. K-Theory*, 6(1):171–196, 2010.
- [34] I.S. Costa. Presentation complexes with the fixed point property. *Geometry and Topology*, 21:1275–1283, 2017.
- [35] I.S. Costa. Puntos fijos de acciones y funciones en 2-complejos. *PhD thesis, Universidad de Buenos Aires*, 2019.
- [36] Rodrigo de Almelda Rosa. *Cálculo de grupos fundamentais de nuvens de pontos através de técnicas de Análise Topológica de Dados*. São Paulo State University - International, Masters thesis, 2021.
- [37] Paul de Lange. The physics & mathematics of microstates in string theory and a monstrous farey tail. *PhD thesis, Univ. van Amsterdam*, 2016.
- [38] Heiko Dietrich, Bettina Eick, and Dörte Feichtenschlager. Investigating p -groups by coclass with GAP. In *Computational group theory and the theory of groups*, volume 470 of *Contemp. Math.*, pages 45–61. Amer. Math. Soc., Providence, RI, 2008.
- [39] Mathieu Dutour Sikirić and Graham Ellis. Wythoff polytopes and low-dimensional homology of Mathieu groups. *J. Algebra*, 322(11):4143–4150, 2009.
- [40] Mathieu Dutour Sikirić, Graham Ellis, and Achill Schürmann. On the integral homology of $\mathrm{PSL}_4(\mathbb{Z})$ and other arithmetic groups. *J. Number Theory*, 131(12):2368–2375, 2011.
- [41] E.G. Minian and K.I. Piterman. The fundamental group of the p -subgroup complex. *arXiv e-prints*, page arXiv:1903.03549, Mar 2019.
- [42] G. Ellis and K. Killeen. Cohomology with local coefficients and knotted manifolds. *Journal Symbolic Computation (to appear)*, 2021.
- [43] G. Ellis and D.H. Kochloukova. Functorial lifts of the non-abelian tensor product and weak commutativity constructions. *Journal of Algebra*, 706:26–78, 2026.

- [44] Graham Ellis. Homological algebra programming. In *Computational group theory and the theory of groups*, volume 470 of *Contemp. Math.*, pages 63–74. Amer. Math. Soc., Providence, RI, 2008.
- [45] Graham Ellis. Cohomological periodicities of crystallographic groups. *J. Algebra*, 445:537–544, 2016.
- [46] Graham Ellis. *An invitation to computational homotopy*. Oxford University Press, Oxford, 2019.
- [47] Graham Ellis and Fintan Hegarty. Computational homotopy of finite regular CW-spaces. *J. Homotopy Relat. Struct.*, 9(1):25–54, 2014.
- [48] Graham Ellis and Simon King. Persistent homology of groups. *J. Group Theory*, 14(4):575–587, 2011.
- [49] Graham Ellis and Luyen Van Le. Homotopy 2-types of low order. *Exp. Math.*, 23(4):383–389, 2014.
- [50] Graham Ellis and Le Van Luyen. Computational homology of n -types. *J. Symbolic Comput.*, 47(11):1309–1317, 2012.
- [51] Graham Ellis and Roman Mikhailov. A colimit of classifying spaces. *Adv. Math.*, 223(6):2097–2113, 2010.
- [52] Graham Ellis, Hamid Mohammadzadeh, and Hamid Tavallaee. Computing covers of Lie algebras. In *Computational group theory and the theory of groups, II*, volume 511 of *Contemp. Math.*, pages 25–31. Amer. Math. Soc., Providence, RI, 2010.
- [53] Graham Ellis and Emil Sköldbberg. The $K(\pi, 1)$ conjecture for a class of Artin groups. *Comment. Math. Helv.*, 85(2):409–415, 2010.
- [54] Graham Ellis and Paul Smith. Computing group cohomology rings from the Lyndon-Hochschild-Serre spectral sequence. *J. Symbolic Comput.*, 46(4):360–370, 2011.
- [55] Dominic Else and Ryan Thorngren. Topological theory of lieb-schultz-mattis theorems in quantum spin systems. *arXiv:1907.08204*, 2019.
- [56] P. Fernández Ascariz. *Módulos cruzados de álgebras conmutativas, homología y HAP*. Teses de doutoramento da Universidade de Santiago de Compostela. Universidad de Santiago de Compostela, Servizo de Publicacións e Intercambio Científico, 2007.
- [57] Matthias R. Gaberdiel, Daniel Persson, Henrik Ronellenfitsch, and Roberto Volpato. Generalized Mathieu Moonshine. *Commun. Number Theory Phys.*, 7(1):145–223, 2013.
- [58] Terry Gannon and Corey Jones. Vanishing of categorical obstructions for permutation orbifolds. *arXiv:1804.08343*, 2018.

- [59] Giles Gardam. Low-dimensional homology of finite coxeter groups. *Zenodo*, <https://zenodo.org/records/4758008>, May 2021.
- [60] Taleea Jalaeeeyan Ghorbanzadeh, Mohsen Parvizi, and Peyman Niroomand. The non-abelian tensor square of p -groups of order p^4 . *Asian-Eur. J. Math.*, 11(6):1850084, 10, 2018.
- [61] Angus Gruen. Computing modular data for drinfeld centers of pointed fusion categories. *Bachelor thesis, The Mathematical Sciences Institute, Australian National University*, 2017.
- [62] Angus Gruen and Scott Morrison. Computing modular data for pointed fusion categories. *Indiana Univ. Math. J.*, 70(2):561–593, 2021.
- [63] Ian Hambleton and Ozgun Unlu. Closed manifold surgery obstructions and the oozing conjecture, 2026.
- [64] S. Hatui. A characterization of finite p -groups by their schur multiplier. *arXiv:1608.06322*, 2016.
- [65] Sumana Hatui. Finite p -groups having Schur multiplier of maximum order. *J. Algebra*, 492:490–497, 2017.
- [66] Sumana Hatui, Vipul Kakkar, and Manoj K. Yadav. The Schur multiplier of groups of order p^5 . *J. Group Theory*, 22(4):647–687, 2019.
- [67] F. Hegarty. *Computational Homology of Cubical and Permutahedral Complexes*. PhD thesis, National University of Ireland Galway. National University of Ireland Galway, 2012.
- [68] Jónathan Heras, Vico Pascual, Ana Romero, and Julio Rubio. Integrating multiple sources to answer questions in algebraic topology. In *Intelligent computer mathematics. 10th international conference, AISC 2010, 17th symposium, Calculemus 2010, and 9th international conference, MKM 2010, Paris, France, July 5–10, 2010. Proceedings*, pages 331–335. Berlin: Springer, 2010.
- [69] Jónathan Heras, Vico Pascual, and Julio Rubio. Using open mathematical documents to interface computer algebra and proof assistant systems. In *Intelligent computer mathematics. 16th symposium, Calculemus 2009, 8th international conference, MKM 2009, held as part of CICM 2009, Grand Bend, Canada, July 6–12, 2009. Proceedings*, pages 467–473. Berlin: Springer, 2009.
- [70] Jónathan Heras, Vico Pascual, and Julio Rubio. A system for computing and reasoning in algebraic topology. In JamesH. Davenport, WilliamM. Farmer, Josef Urban, and Florian Rabe, editors, *Intelligent Computer Mathematics*, volume 6824 of *Lecture Notes in Computer Science*, pages 295–297. Springer Berlin Heidelberg, 2011.

- [71] Rubén A. Hidalgo. Extending finite free actions of surfaces. 2024.
- [72] A. Holzinger. On topological data mining. *Lecture Notes in Computer Science*, 8401:331–356, 2014.
- [73] Andreas Holzinger and Igor Jurisica. Knowledge discovery and data mining in biomedical informatics: The future is in integrative, interactive machine learning solutions. In *Interactive Knowledge Discovery and Data Mining in Biomedical Informatics - State-of-the-Art and Future Challenges*, pages 1–18. 2014.
- [74] A. Hoshi, M.-c. Kang, and A. Yamasaki. Multiplicative invariant fields of dimension ≤ 6 . *arXiv.org*, *arXiv:1609.04142*, pages 1–105, 2016.
- [75] A. Hoshi, M.-c. Kang, and A. Yamasaki. Degree three unramified groups and noether’s problem for groups of order 243. *arXiv.org*, *arXiv:1710.01958*, pages 1–61, 2017.
- [76] Akinari Hoshi, Kazuki Kanai, and Aiichi Yamasaki. Norm one tori and Hasse norm principle. *Math. Comp.*, 91(337):2431–2458, 2022.
- [77] Akinari Hoshi, Kazuki Kanai, and Aiichi Yamasaki. Norm one tori and Hasse norm principle, II: Degree 12 case. *J. Number Theory*, 244:84–110, 2023.
- [78] Akinari Hoshi and Aiichi Yamasaki. Rationality problem for algebraic tori. *Mem. American Math. Soc. (to appear)*, pages 1–146, 2016.
- [79] Akinari Hoshi and Aiichi Yamasaki. Hasse norm principle for metacyclic extensions with trivial schur multiplier, 2025.
- [80] Santanil Jana. Classifying space for commutativity and unordered flag manifolds. *PhD thesis, University of British Columbia*, 2023.
- [81] Santanil Jana. On the cohomology of the total space of classifying space for commutativity in $u(3)$. 2023.
- [82] U. Jezernik. *Universal Commutator Relations*. PhD thesis, University of Ljubljana. University of Ljubljana, 2016.
- [83] Urban Jezernik and Primož Moravec. Bogomolov multipliers of groups of order 128. *Exp. Math.*, 23(2):174–180, 2014.
- [84] T. Johnson-Freyd. The moonshine anomaly. *arXiv:1707.08388*, pages 1–15, 2017.
- [85] T. Johnson-Freyd and D. Treumann. $H^4(Co_0; \mathbb{Z}) = \mathbb{Z}_{24}$. *arXiv:1707.07587v1*, pages 1–14, 2017.
- [86] T. Johnson-Freyd and D. Treumann. Third homology of some sporadic finite groups. *arXiv:1810.00463v1*, pages 1–29, 2018.

- [87] A. Joshi. *MATHIEU MOONSHINE: From M_{24} to M_{12}* . MSc thesis, IIT Madras. 2016.
- [88] Renu Joshi and Siddhartha Sarkar. On the Schur multiplier of finite p -groups of maximal class. *J. Group Theory*, 26(3):533–545, 2023.
- [89] David Joyner. A primer on computational group homology and cohomology using GAP and SAGE. In *Aspects of infinite groups*, volume 1 of *Algebra Discrete Math.*, pages 159–191. World Sci. Publ., Hackensack, NJ, 2008.
- [90] Tomasz Kaczynski and Marian Mrozek. The cubical cohomology ring: an algorithmic approach. *Found. Comput. Math.*, 13(5):789–818, 2013.
- [91] Kazuki Kanai. On the hasse norm principle and the davenport-hasse lifting theorem. *PhD Thesis, Graduate School of Science and Technology, Nigata University*, page 217pp, March 2022.
- [92] M.-C. Kang and B. Kunyavskii. The bogomolov multiplier of rigid finite groups. *Archiv der Mathematik*, pages 209–218, 102 2014.
- [93] Anton Kapustin and Ryan Thorngren. Anomalous discrete symmetries in three dimensions and group cohomology. *Phys. Rev. Lett.*, 112:231602, Jun 2014.
- [94] Daniel Kasprzycki, John Nicholson, and Simona Vesela. Stable equivalence relations on 4-manifolds, 2024.
- [95] E. Khamseh and S. Hadi Jafari. Computing the 2-nilpotent multiplier of 2-generator p -groups of class 2. *Indian J Pure Appl Math*, 2024.
- [96] Xianghong Kong, Xingsi Liu, Shuihua Yang, Zhiyuan Yan, Weijin Chen, Zhixia Xu, and Cheng-Wei Qiu. Nearest-neighbor tight-binding realization of hyperbolic lattices with $\mathbb{Z}_2$ gauge structures, 2025.
- [97] V. Kurlin. Computing invariants of knotted graphs given by sequences of points in 3-dimensional space. *Topological methods in data analysis and visualization IV : theory, algorithms, and applications. Mathematics and visualization. (Springer)*, 2017.
- [98] Larry A. Lambe. An algebraic study of the Klein bottle. *J. Homotopy Relat. Struct.*, 11(4):885–891, 2016.
- [99] Jong Yeon Lee. Fractionalization, emergent gauge dynamics, and topology in quantum matter. 2020.
- [100] J.Y. Lee, A. Turner, and A. Vishwanaith. Building symmetry enriched topological phases from a bipartite lattice construction and anyon condensation. *arXiv e-prints*, page arXiv:1802.02155, Feb 2018.

- [101] Linfeng Li. Index relations and fusion rules: Explorations of supersymmetric, conformal, and topological field theories. *PhD Thesis, Queen Mary University of London*, page 232pp, 15 August 2021.
- [102] Linfeng Li. Index relations and fusion rules: Explorations of supersymmetric, conformal, and topological field theories. *Queen Mary, University of London, PhD thesis*, 2021.
- [103] Chunxiao Liu and Weicheng Ye. Crystallography, group cohomology, and lieb-schultz-mattis constraints, 2024.
- [104] Pedro Lopes and Manpreet Singh. Constructing an infinite family of quandles from a quandle. 2022.
- [105] Rafał Lutowski and Bartosz Putrycz. Spin structures on flat manifolds. *J. Algebra*, 436:277–291, 2015.
- [106] Mohamad Maassarani. Isoclinic groups and conjugacy quandles, 2026.
- [107] Pablo Magni. Derived equivalences of generalized kummer varieties. 2023.
- [108] Kimball Martin and Nahid Walji. Distinguishing finite group characters and refined local-global phenomena. *arXiv*, 2016.
- [109] Kirk Mcdermott. On $\mathbf{u}$ -substitutions for group presentations, 2026.
- [110] I. Michailov, I. Ivanov, A. Aleksandrina, and G. Llianova. Algorithmic determination of isoclinism classes for 6-generator groups of nilpotency class 2. *Proceedings of the University of Ruse Conference, 2015*. <http://conf.uni-ruse.bg/bg/docs/cp15/6.1/6.1-4.pdf>, pages 24–27, 2015.
- [111] I. Michailov, I. Ivanov, and N. Ziapkov. Algorithmic generation of isoclinism classes for 4-generator groups of nilpotency class 2. *Proceedings of the University of Ruse Conference, 2015*. <http://conf.uni-ruse.bg/bg/docs/cp15/6.1/6.1-3.pdf>, pages 24–27, 2015.
- [112] Pascal Michel. Homology of groups and third busy beaver function. *Internat. J. Algebra Comput.*, 20(6):769–791, 2010.
- [113] M. Mignard and P. Schauenburg. Morita equivalence of pointed fusion categories of small rank. *HAL Id: hal-01573708*, pages 1–46, 2017.
- [114] Roman Mikhailov and Jie Wu. On homotopy groups of the suspended classifying spaces. *Algebr. Geom. Topol.*, 10(1):565–625, 2010.
- [115] Thomas Titz Mite and Stefan Witzel. A c_2 -tilde-lattice that is not residually finite. 2023.

- [116] Yasushi Mizusawa. On certain 2-extensions of $\mathbb{Q}$ unramified at 2 and ∞ . *Osaka J. Math.*, 53(4):1063–1088, 10 2016.
- [117] Yasushi Mizusawa. On various pro- p -extensions of number fields with restricted ramification — a survey. *Kyoto University Research Information Repository 400: Research Institute for Mathematical Sciences*, B86, 7 2021.
- [118] Mohammad Reza R. Moghaddam and Peyman Niroomand. Some properties of certain subgroups of tensor squares of p -groups. *Comm. Algebra*, 40(3):1188–1193, 2012.
- [119] Primož Moravec. Absence of twisting for non-trivial discrete torsion, 2025.
- [120] M. Muller Lopes Rocha. *Ejemplos de algebras de Hopf semisimples y de algebras de Hopf con la propiedad de Chevalley dual*. PhD thesis, Universidad Nacional de Cordoba. University Universidad Nacional de Cordoba, 2016.
- [121] D. Müllner. *Orientation Reversal of Manifolds*. Bonner mathematische Schriften. D. Müllner, 2008.
- [122] Paul Mücksch and Masahiko Yoshinaga. Milnor fibrations and oriented matroids, 2026.
- [123] Neha Nanda, Mahender Singh, and Manpreet Singh. Knot invariants from derivations of quandles. *arXiv:1804.01113*, 2018.
- [124] Afifi Nawi, Nor Ali, Nor Muhainiah Mohd, and Nor Haniza Sarmin. Aip conference proceedings 2266, 060002 (2020); <https://doi.org/10.1063/5.00183402266>, 060002© 2020 author(s).the computation of schur multiplier and capability of pairs of groups of order p^4 . *AIP Conference Proceedings 2266*, 2020.
- [125] Georges Neaime. Interval structures, hecke algebras, and krammer’s representations for the complex braid groups $b(e,e,n)$, 2018.
- [126] P. Niroomand and R. Rezaei. On the exterior degree of finite groups. *Comm. Algebra*, 39(1):335–343, 2011.
- [127] Peyman Niroomand and Rashid Rezaei. The exterior degree of a pair of finite groups. *Mediterr. J. Math.*, 10(3):1195–1206, 2013.
- [128] Peyman Niroomand and Francesco G. Russo. Probabilistic properties of the relative tensor degree of finite groups. *Indagationes Mathematicae*, 27(1):147–159, 2016.
- [129] A. Odabaş, E.Ö. Uslu, and E. Ilgaz. Isoclinism of crossed modules. *J. Symbolic Comput.*, 74:408–424, 2016.
- [130] Alper Odabas. Exterior isoclinism of crossed modules. 2022.

- [131] Yunqing Ouyang, Qing-Rui Wang, Zheng-Cheng Gu, and Yang Qi. Computing classification of interacting fermionic symmetry-protected topological phases using topological invariants*. *Chinese Physics Letters*, 38(12):127101, dec 2021.
- [132] Nansen Petrosyan and Bartosz Putrycz. On cohomology of crystallographic groups with cyclic holonomy of split type. *J. Algebra*, 367:237–246, 2012.
- [133] Dung Phuong Phan, Tuan Anh Bui, and Alexander D. Rahm. Computations regarding the torsion homology of oeljeklaus-toma manifolds. 2024.
- [134] Dung Phuong Phan, Tuan Anh Bui, and Alexander D. Rahm. Studies of the torsion in the homology of Oeljeklaus-Toma manifolds. *Differential Geom. Appl.*, 101:Paper No. 102304, 20, 2025.
- [135] R. Radhakrishnan. Algebraic structure of topological and conformal field theories. *PhD thesis, Queen Mary, University of London*, Dec 2022.
- [136] Alexander D. Rahm. The homological torsion of PSL_2 of the imaginary quadratic integers. *Trans. Amer. Math. Soc.*, 365(3):1603–1635, 2013.
- [137] Alexander D. Rahm. Accessing the cohomology of discrete groups above their virtual cohomological dimension. *J. Algebra*, 404:152–175, 2014.
- [138] Alexander D. Rahm. The subgroup measuring the defect of the abelianization of $\mathrm{SL}_2(\mathbb{Z}[i])$. *J. Homotopy Relat. Struct.*, 9(2):257–262, 2014.
- [139] Alexander D. Rahm, Anh Tuan Bui, and Matthias Wendt. The 2-torsion in the Farrell-Tate cohomology of $\mathrm{PSL}_4(\mathbb{Z})$, and torsion subcomplex reduction via discrete Morse theory. *J. Pure Appl. Algebra*, 230(4):Paper No. 108225, 2026.
- [140] Pradeep K. Rai and Manoj K. Yadav. On Sh-rigidity of groups of order p^6 . *J. Algebra*, 428:26–42, 2015.
- [141] Pradeep Kumar Rai. Automorphisms of groups. *PhD Thesis, Homi Bhabha National Institute*, page 116pp, 2014.
- [142] J. Roberts. *ALGORITHMS FOR UPPER BOUNDS OF LOW DIMENSIONAL GROUP HOMOLOGY*. PhD thesis, University of Kentucky. University of Kentucky, 2010.
- [143] Joshua Roberts. An algorithm for low dimensional group homology. *Homology, Homotopy Appl.*, 12(1):27–37, 2010.
- [144] Marc Röder. Geometric algorithms for resolutions for Bieberbach groups. In *Computational group theory and the theory of groups, II*, volume 511 of *Contemp. Math.*, pages 167–178. Amer. Math. Soc., Providence, RI, 2010.

- [145] Ana Romero, Graham Ellis, and Julio Rubio. Interoperating between computer algebra systems: Computing homology of groups with kenzo and gap. In *Proceedings of the 2009 International Symposium on Symbolic and Algebraic Computation*, ISSAC '09, pages 303–310, New York, NY, USA, 2009. ACM.
- [146] Ana Romero and Julio Rubio. Computing the homology of groups: the geometric way. *J. Symbolic Comput.*, 47(7):752–770, 2012.
- [147] Ana Romero and Julio Rubio. Homotopy groups of suspended classifying spaces: an experimental approach. *Math. Comp.*, 82(284):2237–2244, 2013.
- [148] Eric Samperton and Carlos Segovia. Topological perspectives on the vanishing of some bogomolov multipliers, 2026.
- [149] Peter Schauenburg. Erratum to: “A higher Frobenius-Schur indicator formula for group-theoretical fusion categories”. *Commun. Math. Phys.*, 350(2):893–896, 2017.
- [150] S. Schönnenbeck. Homologiegruppen von Einheitengruppen von Ordnungen. *Masterarbeit, RWTH Aachen*, September 2013.
- [151] S. Schönnenbeck. Hecke operators for algebraic modular forms . *Thesis, RWTH Aachen*, September 2016.
- [152] S. Schönnenbeck. Resolutions for unit groups of orders. *ArXiv e-prints*, September 2016.
- [153] M. Schwald. On the definition of irreducible holomorphic symplectic manifolds and their singular analogs. *arXiv e-prints*, page arXiv:2003.06004, Dec 2020.
- [154] Sarvesh Srinivasan, Jian-Hao Zhang, Yang Qi, and Zhen Bi. Classification of average crystalline topological superconductors through a generalized real-space construction, 2025.
- [155] Dennis Sweeney. Homotopy types of small semigroups, 2026.
- [156] Markus Szymik. The third Milgram–Priddy class lifts. *arxiv.org/pdf/1909.05064.pdf*, pages 1–6, 2019.
- [157] Moe Tabei. hap-crosschecks (github package), 2026.
- [158] Naman Taggar. Simplicialhomology: implementation of abstract simplicial complex in mathematica, 2026.
- [159] Raphaël Tinarrage. Simplicial approximation to cw complexes with spherical delaunay triangulations. volume 367, 2026. Cited by: 0.

- [160] Bui Anh Tuan and Graham Ellis. The homology of $SL_2(\mathbb{Z}[1/m])$ for small m . *J. Algebra*, 408:102–108, 2014.
- [161] Abraham Rojas Vega. The schur multiplier of $\mathfrak{sl}_2$ over finite commutative rings, 2026.
- [162] Lukas Vokřínek. Algorithmic aspects of topological problems. *Habilitation thesis, Brno University*, 2017.
- [163] Kelly Wang. Periodic families in the homology of $gl_n(f_2)$, 2026.
- [164] Alison Warman, Yuhan Gai, and Sakura Schafer-Nameki. Twin phases: Intrinsic deconfined quantum criticality, 2026.
- [165] Zhengyao Wu. Rationality of the center of a generic division algebra with a group action over a field of characteristic 0. <https://arxiv.com/papers/view/2606.0006/>, 2026.
- [166] Jian-Hao Zhang and Shuo Yang. Tensor network representations of fermionic crystalline topological phases on two-dimensional lattices. 2021.
- [167] Wayne Zheng, Jia-Wei Mei, and Yang Qi. Classification and monte carlo study of symmetric Z_2 spin liquids on the triangular lattice. *arXiv:1505.05351*, 2015.
- [168] Alexander Ziegler. The chow ring of the 2-sylow subgroup of $GL(4, 2)$, 2025.
- [169] V. Álvarez, J.A. Armario, M.D. Frau, and P. Real. A mathematica notebook for computing the homology of iterated products of groups. In Andres Iglesias and Nobuki Takayama, editors, *Mathematical Software - ICMS 2006*, volume 4151 of *Lecture Notes in Computer Science*, pages 47–57. Springer Berlin Heidelberg, 2006.